Estuaries and Coasts Vol. 30, No. 5, p. 869–877 October 2007

Salt Marsh Vegetation Recovery on the Bay of Fundy

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ABSTRACT: Beginning in the 17th century many Bay of Fundy marshes were diked and drained for agricultural use, but storm breaching of dikes and failure of tidal gates has returned tidal flooding to some marshes. We investigated two breached and undiked salt marsh pairs in the Bay of Fundy to assess the potential for recovery of these reclaimed lands and improve our knowledge about salt marsh restoration. We examined the distribution of major species with respect to elevation and compared plant cover and production of reference and recovering marshes. The vertical range of both *Spartina alterniflora* and *Spartina patens* increased with tidal range – a condition recognized for *S. alterniflora*, but not previously reported for *S. patens*. Our results reveal that *S. alterniflora* and *S. patens* are inundated less frequently than in microtidal marshes and tolerate a large variation in inundation frequency and depth. The upper portions of the *S. patens* zone may not be submerged by tidal waters in some years, so hydroperiod (and its associated stresses) may play less of a role in limiting the lower elevation of a species' distribution in the high marsh. Our comparisons of reference and recovering diked marshes inform not only restoration activities on the Bay of Fundy, but also give a perspective on recovery trajectories of reclaimed salt marshes in general. Regionally, the broad elevation range of *S. alterniflora* and high sediment deposition rates impart Bay of Fundy marshes with a high resilience and prospects for success of deliberate restoration efforts are promising, without the addition of fill required for marsh restoration in some regions. While marsh loss occurs in other areas the Bay of Fundy provides an opportunity to regain this resource.

Introduction

Reclamation, or conversion of tidal salt marshes to agricultural land, has occurred through diking and draining activities on many coasts, including those of The Netherlands, France (Butzer 2002), Spain (Onaindia and Amezaga 1999), United Kingdom (Pethick 2002), the Northwest United States (e.g., Thom et al. 2002), California (e.g., Williams and Orr 2002), and Quebec's St. Lawrence River estuary (Hatvany 2003). On the Bay of Fundy salt marsh conversion began in the 17th century and estimates of marsh loss are as high as 85% (Ganong 1903). Efforts to restore marshes are growing internationally, particularly in England and the Northwest U.S. There is growing interest to restore former Bay of Fundy salt marshes by removing dikes and other tidal barriers (Harvey 2000; Gulf of Maine Council Habitat Restoration Subcommittee 2004), but limited experience to draw upon. We can find analogues for future restoration projects at sites recovering with no or minimal assistance. This study investigates the recovery of vegetation at two recovering and reference salt marsh pairs in the Bay of Fundy. Our results expand our knowledge of the range of conditions where salt marsh growth is possible, directly inform future restoration efforts on the Bay of Fundy, and provide a broader perspective on recovery trajectories of reclaimed marshes.

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When tidal salt marshes are diked the loss of tidal flooding and the sediments it brings result in decreased soil salinity (Ganong 1903; Daiber 1986; French 2006) and relative elevation (Roman et al. 1984; Allen 1999; Crooks and Pye 2000; Weinstein and Weishar 2002). Salts are eventually leached from the sediment by precipitation. Lower water tables enhance sediment compaction due to dewatering and increased rates of decomposition. Loss of tidally derived sediments prevents vertical accretion. Even without the continued increase in eustatic sea level these systems would decline in relative elevation.

A breach in the dike can be created to restore or reactivate salt marshes (Blott and Pye 2004) or unintentional dike breaches can occur during storm events and allow marsh recovery to proceed with little or no human influence (Crooks et al. 2002). Investigations of these recovering sites can offer long-term information about vegetation and geomorphology (e.g., Crooks et al. 2002; French 2006) and provide insight into what factors could constrain restoration at other sites.

A means to assess progress or success of restoration is by determination of similarity of vegetation cover and biomass between reference and restored sites (Warren et al. 2002). In this study we compare the vegetation at two recovering and two reference marshes to assess the progress of recovery. Progress is assessed by calculation of similarity indices based on plant cover and end-of-season standing crop of each marsh pair. Use of similarity indices enables a comparison of recovery rates in Fundy marshes to those in other regions.

Material and Methods

Study Areas

The Bay of Fundy, situated between New Brunswick and Nova Scotia, has a spring tidal range that increases from 5 m at the mouth to > 16 m in the upper reaches (Canadian Hydrographic Survey 2005). Vegetation at reference (undiked) and recovering (storm-breached dike) marsh pairs in both the lower and upper Bay was sampled to reflect this gradient in tidal range (Conner et al. 2001). The Dipper Harbour marsh (10.8 ha: 45°06′N, 66°26′W) is the outer Bay reference site because it has never been diked or ditched while the Saint's Rest marsh (94.7 ha: 45°13′N, 66°07′W) was diked sometime between 1786 and 1864 and is the recovering site in the outer Bay. The tidal range of the Dipper Harbour marsh is 6 m at mean tide and 8 m at large tides, and slightly larger at the Saint's Rest marsh, 6.7 and 8.9 m, respectively. Tidal flooding was returned to Saint's Rest marsh when the dike was breached in the 1950s (Noel et al. 2005) and later portions were removed to enhance tidal exchange. The Wood Point marsh (16.9 ha: 45°52′N, 64°21′W) and John Lusby marsh (600 ha: 45°47′N, 64°17′W) are the two upper Bay sites situated in the Cumberland Basin. The tidal range at the Wood Point marsh is 10.5 m at mean tides and 14.7 m at large tides, but slightly larger at the John Lusby marsh, 11.0 and 15.1 m, respectively. The Wood Point marsh, also referred to as Allen Creek marsh, was not diked and serves as the reference site. The John Lusby marsh was diked sometime between 1686 and 1693 (Clarke 1968) and the dikes failed in 1947 (Graf 2004).

Vegetation

Two types of vegetation plots were established in each marsh: cover plots and end-of-season standing crop plots. Triplicate 0.25-m^2 standing crop plots were established in July 2005 at the Wood Point (n = 36) and John Lusby (n = 36) marshes. We excluded areas behind remnant dikes that were receiving restricted tidal flow at the John Lusby marsh. The percent vegetation cover was described in August 2005 at a 1-m^2 plot surrounding piezometers stationed along transects established for studies of subsurface hydrology (Byers 2006). The total cover always equalled 100%.

Several data sets were employed to characterize elevation and channel locations, and locate standing crop plots at varying elevations and distances. Transects across the four marsh platforms were recorded with a differential global positioning system (DGPS). For the Wood Point marsh, additional DGPS data collected from previous field campaigns by van Proosdij (2001) and van Proosdij et al. (2006) were used. The elevations of the bayward edge of vegetation and the vegetated edge along primary tidal channels (defined as channels first flooded by incoming flood waters) were measured at the approximate transition between vegetation and mudflat (i.e., the lowest edge of vegetation). The elevation of vegetated cliff edges at Wood Point and John Lusby marshes was measured at points where vegetation terminated at the top of marsh cliffs.

Vegetation was harvested from 72 plots at the Wood Point and John Lusby marshes in late August 2005 when peak standing crop occurred. Unfortunately, some grass species had already senesced by this time. Lack of seed heads in these specimens made it impossible to make unequivocal identifications. Samples were washed until the rinse water ran clear and the remaining sediment lodged in *S. alterniflora* ligules and litter was removed by hand. Samples were sorted by species, air dried for 3 mo, then weighed.

Elevation and Tidal Inundation

The elevation relative to mean sea level (MSL) was measured with a DGPS and referenced to the Canadian Geographic Vertical Datum 1928 (CGVD28). The level of mean high water (MHW) and higher high water (HHW) for each marsh were determined using methods described in Canadian Tide and Current Tables (Canadian Hydrographic Survey 2005) and additional calculations. Because mean sea level is an intrinsically different measure of water level compared to MHW and HHW, it is important to consider elevations relative to both types of measurements.

The MHW and HHW are referenced to chart datum (i.e., a plane below which the tide will seldom fall) and these values were converted to CGVD28 to allow their comparison with our DGPS-derived elevations. This conversion involves subtracting mean water level from both the mean tide level and large tide level (Webster et al. 2004). Values for the Saint's Rest marsh were based on the Saint John reference port, the Dipper Harbour marsh was based on the secondary port of Dipper Harbour West, and the two upper Bay marshes were based on the secondary port of Pecks Point. The Wood Point and John Lusby marshes are located 9 and 22 km up-Bay from Pecks Point, respectively, requiring additional corrections to accommodate the 0.017 m km^{-1} increase in MHW and 0.029 m km^{-1} increase in HHW towards the head of the Cumberland Basin (Gordon et al. 1985).

We investigated the relationships among zonation of dominant species, elevation relative to MSL and tide levels, frequency of tidal inundation, and depth of tidal inundation. Only monospecies plots, defined as plots in which 95% or more of the cover or end-of-season standing crop consisted of one species, were considered when determining species' elevations and tidal inundation. The frequency and depth of inundation were calculated by comparing predicted elevations (relative to MSL) of high tides at Saint John for 2005, a low point in the 18.6-yr tidal cycle, and 1995, a year with a large number of extreme tides, to elevations relative to MSL of various vegetated features and monospecies plots. We assumed that a given elevation was flooded when tide elevation equalled or exceeded this elevation.

Statistical Analyses

We used a modified Sorensen's K index (Bray and Curtis 1957; West 1966) to estimate similarity in species composition, species cover, and species productivity between sites. Plots containing *Juncus gerardii* and *Carex palacea* were excluded from this analysis because our study areas at Saint's Rest and John Lusby marshes did not include this upland edge zone. An unweighted similarity index was calculated: Similarity = $(2A/[2A + B + C]) \times 100\%$, where A is the number of species in common between the two sites, B is the number of species exclusive to the first site, and C is the number of species exclusive to the second site. The term $2A$ was added to the denominator so that the resulting value was between 0 and 100 (Thom et al. 2002). A weighted Sorensen index of similarity was calculated using cover or end-of-season standing crop. A is the sum of the lower values (cover or end-of-season standing crop) for each species that the two sites have in common, B is the species and summed values (cover or end-of-season standing crop) exclusive to the first site, and C is the species and summed values (cover or end-of-season standing crop) exclusive to the second site.

Results

Tidal Range, Elevation, and Tidal Inundation

The tidal range, MHW, and HHW levels increase up-Bay (Fig. 1). The Dipper Harbour marsh has the lowest levels because it is closest to the mouth of the Bay, and John Lusby marsh has the highest levels because it is closest to the head of the Bay. The levels of MHW and HHW increase 0.3 m each over the 28 km distance between Dipper Harbour marsh and Saint's Rest marsh. The Wood Point and John Lusby marshes are 13 km apart and MHW and HHW levels increase by 0.2 and 0.4 m, respectively, over this distance.

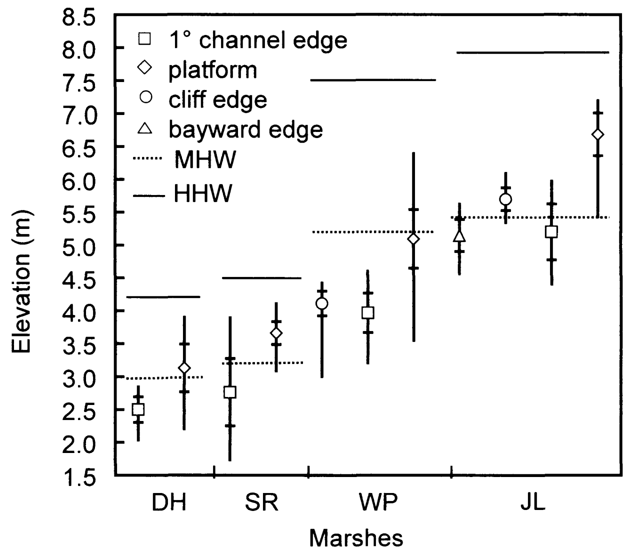

Fig. 1. Elevation relative to mean sea level (referenced to CGVD28) of vegetated features at marsh study sites. Symbols represent mean elevation, vertical lines represent elevation range, and tick marks represent ± 1 SD. Horizontal dotted lines represent mean high water (MHW) and horizontal solid lines represent higher high water (HHW). Primary channel edge and bayward edge refer to the approximate transition between vegetation and mudflat (i.e., the lowest edge of vegetation). Cliff edge refers to vegetation termination at the top of marsh cliffs. DH = Dipper Harbour marsh, SR = Saint's Rest marsh, WP = Wood Point marsh, JL = John Lusby marsh.

Differences in elevation among vegetated features (Fig. 1) and selected plant species (Fig. 2) among the four Bay of Fundy marshes are striking. Increases in elevation relative to MSL up-Bay are explained by increasing tidal ranges. Consider the mean marsh platform elevation. The Dipper Harbour and Saint's Rest marshes have mean marsh platform elevations near MHW, 3.13 and 3.65 m, respectively. On average the Dipper Harbour marsh platform is 0.1 m below MHW, while the Saint's Rest marsh platform is 0.6 m above MHW. The Wood Point and John Lusby marshes have mean marsh platform elevations of 5.09 and 6.08 m, respectively, which are 1.5 to 3.6 m higher than the two lower Bay marshes. While the platform at Wood Point marsh is only 0.1 m below MHW, the platform at John Lusby marsh is 1.3 m higher than MHW. The edge of vegetation along primary channels at the Wood Point marsh averages 1.2 m below MHW and at the John Lusby marsh is 0.2 m below MHW.

Although both upper Bay marshes have cliffs along the Bay, the Wood Point marsh cliff is vegetated by *Spartina alterniflora*, while the cliff at John Lusby marsh is vegetated mainly by *Spartina patens*. The cliff vegetation differs between these two marshes because, on average, the cliff elevation is 1.1 m below MHW at the Wood Point marsh and at

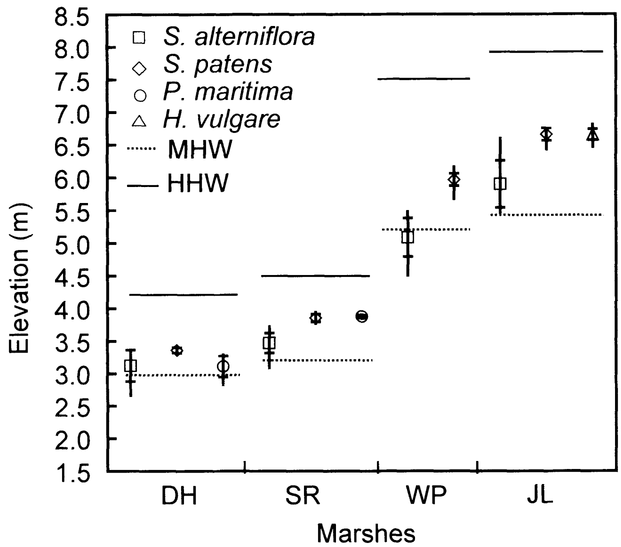

Fig. 2. Elevation relative to mean sea level (referenced to CGVD28) of selected species in monospecies plots at marsh study sites. Symbols represent mean elevation, vertical lines represent elevation range, and tick marks represent ± 1 SD. Horizontal dotted lines represent mean high water (MHW) and horizontal solid lines represent higher high water (HHW). DH = Dipper Harbour marsh, SR = Saint's Rest marsh, WP = Wood Point marsh, JL = John Lusby marsh.

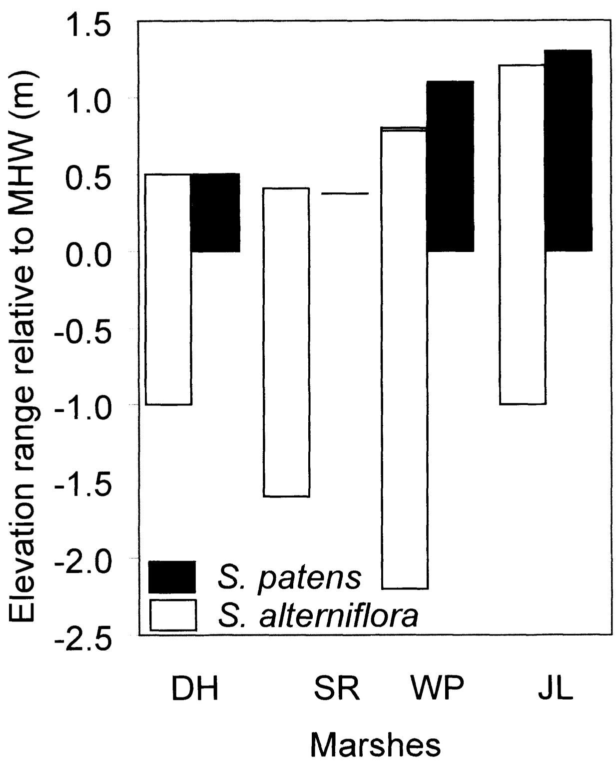

Fig. 3. Elevation range of *Spartina alterniflora* and *Spartina patens* relative to mean high water (MHW). These ranges are based on data from all vegetation plots (i.e., both monospecies and mixed plots) and DGPS surveys of vegetated features. Only the lower edge of *S. patens* was sampled at Saint's Rest and it is depicted with a dash.

the John Lusby marsh the cliff lies, on average, at 0.3 m above MHW (Fig. 1). *S. alterniflora* is present below the cliff at the John Lusby marsh and extends towards the Bay - the bayward edge of this vegetation lies at 0.3 m below MHW, on average.

As expected, *S. patens* occurred at higher elevations than *S. alterniflora* within the monospecies plots, but their occurrence did not correspond to a consistent elevation relative to either HHW or MHW (Fig. 2). Figure 3 displays the vertical range of *S. alterniflora* and *S. patens*, species for which the most complete data is available, at all four marshes relative to MHW. The data suggest that the vertical range of both species increases with increasing tidal range. *S. alterniflora* at the John Lusby marsh is an exception – it occurs in a narrower vertical range. The minimum elevation of occurrence for *S. alterniflora* decreases with tidal range while the maximum elevation of occurrence generally increases with tidal range, again the minimum elevation at the John Lusby marsh is an exception to this trend. While the maximum elevation of *S. patens* also follows this trend, the minimum elevation does not.

Inundation frequency of monospecies plots is higher at the Dipper Harbour marsh than the Saint's Rest marsh (Table 1). On average, flooding frequency of *S. patens* is lowest at the John Lusby marsh. Frequency and depth of flooding are often greater in 1995 – a year with a large number of extreme tides (Table 1).

Vegetation

The mean plant cover and end-of-season standing crop by marsh vary between reference and recovering marshes (Tables 2 and 3). The plant cover at the Dipper Harbour marsh is roughly divided among *S. alterniflora, Plantago maritima,* and *S. patens.* The Saint's Rest marsh is dominated by *S. alterniflora* (86.3%). *S. alterniflora* dominates at the Wood Point marsh, but there is also a high percentage of *S. patens* and some *Puccinellia* spp. Although *S. alterniflora* occurs at the John Lusby marsh, this marsh is characterized by high marsh vegetation including an unidentified grass species (U1), *S. patens,* and *Puccinellia* spp.

End-of-season standing crop was only measured at the upper Bay sites and it paralleled the pattern in cover (Table 3). End-of-season standing crop is roughly split between *S. alterniflora* and *S. patens* at the Wood Point marsh plots, while *S. patens* and *Hordeum vulgare* dominate these samples at the John Lusby marsh. Other high marsh species, which have

TABLE 1. Inundation frequency per year and depth across various vegetated features and monospecies plots based on predicted high tides in 2005 and 1995. Numbers denote mean (maximum, minimum). 1° channel edge and bayward edge refer to the approximate transition between vegetation and mudflat (i.e., the lowest edge of vegetation). Sa = *Spartina alterniflora*, Sp = *Spartina patens*, Hv = *Hordeum vulgare*, and Pm = *Plantago maritima*. Depths of 0 m indicate that features or species were inundated less than 0.1 m because tidal elevations were reported to the nearest decimeter.

Feature/Species	Inundation frequency/yr 2005	Inundation frequency/yr 1995	Inundation depth (m) 2005	Inundation depth (m) 1995
Dipper Harbour (45°5′N, 66°26′W)				
1° channel edge	446 (661, 206)	534 (693, 348)	0.4 (0.7, 0.4)	0.5 (0.8, 0.3)
Sa	128 (325, 70)	186 (430, 80)	0.3 (0.3, 0.2)	0.2 (0.4, 0.2)
Sp	46 (70, 46)	53 (80, 53)	0.1 (0.2, 0.1)	0.1 (0.2, 0.1)
Pm	128 (206, 70)	186 (348, 80)	0.2 (0.4, 0.2)	0.2 (0.3, 0.2)
Saint's Rest(45°13′N, 66°7′W)				
1° channel edge	446 (705, 18)	571 (704, 28)	0.5 (1.3, 0.2)	0.5 (1.5, 0.1)
Sa	100 (258, 70)	171 (408, 80)	0.3 (0.4, 0.2)	0.2 (0.4, 0.2)
Sp	18 (33, 18)	28 (46, 28)	0.2 (0.2, 0.2)	0.0 (0.1, 0.0)
Pm	18 (33, 18)	28 (46, 28)	0.2 (0.2, 0.2)	0.0 (0.1, 0.0)
Wood Point (45° 50′N, 64° 22′W)				
cliff edge	701 (705, 661)	704 (704, 693)	0.9 (2.0, 0.7)	1.1 (2.2, 0.8)
1° channel edge	705 (705, 589)	704 (704, 648)	1.0 (1.8, 0.6)	1.2 (2.2, 0.7)
Sa	325 (631, 128)	430 (685, 186)	0.3 (0.6, 0.2)	0.4 (0.7, 0.2)
Sp	18 (70, 2)	17 (80, 0)	0.1 (0.2, 0.0)	0.1 (0.2, 0.0)
John Lusby (45° 48′N, 56° 16′W)				
bayward edge	631 (705, 401)	685 (704, 485)	0.6 (1.0, 0.4)	0.7 (1.2, 0.4)
cliff edge	325 (549, 128)	430 (617, 186)	0.3 (0.5, 0.2)	0.4 (0.6, 0.2)
1° channel edge	589 (705, 158)	648 (704, 241)	0.6 (1.2, 0.3)	0.7 (1.4, 0.3)
Sa	206 (446, 18)	293 (534, 17)	0.3 (0.4, 0.1)	0.3 (0.5, 0.1)
Sp	10 (46, 2)	1 (53, 0)	0.0 (0.2, 0.0)	0.0 (0.1, 0.0)
Hv	10 (33, 2)	1 (35, 0)	0.0 (0.1, 0.0)	0.0 (0.1, 0.0)

greater end-of-season standing crop at the John Lusby marsh compared to the Wood Point marsh, include *Puccinellia* spp., *Hordeum jubatum*, *Limonium nashii*, and an unidentified grass (U1).

We used cover and end-of-season standing crop to calculate similarity indices comparing reference and recovering marshes (Fig. 4). The unweighted similarity of the Dipper Harbour marsh and Saints Rest marsh cover plots is high because both marshes have many species in common; only five species were absent from either marsh. *Spergularia canadensis*, *Puccinellia* spp., and *Glaux maritima* are only present in plots sampled at the Dipper Harbour marsh, while *Festuca rubra* and *Atriplex patula* are only present in plots sampled at the Saint's Rest marsh. The weighted similarity of the Dipper

TABLE 2. Mean percent cover (± 1 standard error) of all plots at each Bay of Fundy marsh.

	Dipper Harbour (n = 24)	Saint's Rest (n = 23)	Wood Point (n = 30)	John Lusby (n = 22)
Spartina alterniflora	28.7 (7.9)	86.3 (4.9)	47.3 (8.2)	13.6 (7.5)
Spartina patens	23.9 (5.5)	4.1 (4.1)	37.5 (7.9)	26.5 (8.9)
Puccinellia spp.	0.2 (0.2)		3.5 (1.4)	23.9 (8.4)
Plantago maritima	23.4 (5.0)	1.7 (1.0)	0.7 (0.5)	
Bare ground	4.0 (2.2)	2.6 (1.8)	5.0 (2.1)	0.5 (0.5)
Juncus gerardii	4.9 (2.4)		3.3 (3.3)	
Carex palaecea	3.5 (2.5)		2.3 (2.3)	
Unidentified 1				27.0 (8.8)
Triglochin maritima	1.1 (0.6)	1.0 (0.6)		2.5 (1.5)
Glaux maritima	6.6 (2.6)			1.4 (1.0)
Sueda maritima	1.6 (1.1)	1.6 (0.7)		
Salicornia europea	0.7 (0.4)	1.4 (0.7)	0.1 (0.0)	
Limonium nashii	1.2 (0.6)	<0.05	<0.05	
Hordeum vulgare				2.3 (2.3)
Unidentified 2				2.3 (2.3)
Festuca rubra		1.3 (1.3)		
Rock	0.3 (0.3)			
Atriplex patula		<0.05	<0.05	
Spergularia canadensis	<0.05			

TABLE 3. Mean (± 1 standard error) end-of-season standing crop (g dry wt m^{-2}) from all plots at each upper Bay of Fundy marsh.

	Wood Point (n = 36)	John Lusby (n = 36)
Spartina alterniflora	190.1 (29.7)	76.5 (27.7)
Spartina patens	159.3 (39.3)	178.1 (30.8)
Hordeum vulgare		112.0 (43.9)
Unidentified 1	4.0 (1.8)	7.4 (2.1)
Limonium nashii	1.2 (1.2)	4.1 (2.4)
Hordeum jubatum		5.7 (2.3)
Pucinellia spp.	0.5 (0.4)	5.3 (2.3)
Unidentified 2		5.3 (3.2)
Salicornia europea	0.7 (0.3)	
Unidentified 3		0.6 (0.4)
Atriplex patula	0.4 (0.4)	0.4 (0.2)
Sueda maritima	0.1 (0.1)	
Triglochin maritima	0.4 (0.4)	
Unidentified 4		<0.05

Harbour and Saint's Rest marshes is closer because the percent cover contributed by species with variable presence is quite low (194%). The unweighted similarity based on cover plots is lower at the Wood Point and John Lusby marshes because these marshes have less species in common; nine species were absent from either marsh. *P. maritima*, *A. patula*, *L. nashii*, and *Salicornia europaea* were only present in the Wood Point marsh vegetation cover plots. *Triglochin maritima*, *G. maritima*, *H. vulgare*, and unidentified grass species (U1 and U2) were present only in the John Lusby marsh vegetation cover plots. The unweighted similarity of the end-of-season standing crop plots at the Wood Point and John Lusby marshes was higher than the cover plots because there were more species in common and only eight species were absent from either marsh.

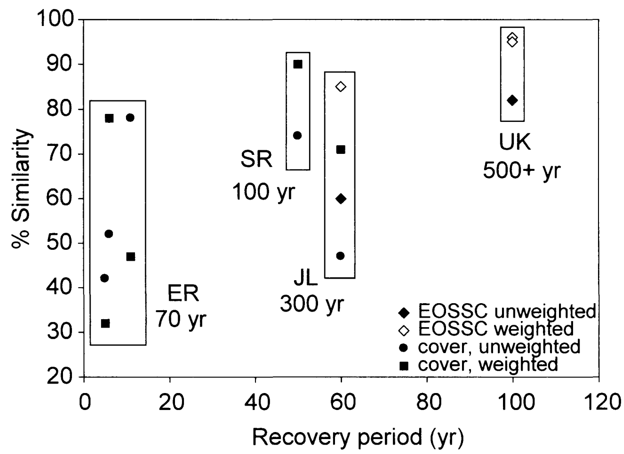

Fig. 4. Weighted and unweighted similarity between reference marshes and sites recovering for various periods. Similarity measures are based upon cover or end-of-season standing crop (EOSSC) as calculated by Thom et al. (2002) for Elk River (ER) in the U.S.; by Crooks et al. (2002) for Northey Island and North Fambridge marshes in the UK; and by this study for Saint's Rest (SR) and John Lusby (JL) marshes. Similarity for the UK marshes was calculated using data provided in their Table 2 of Crooks et al. Similarity measures from the same sites are indicated by a box labelled with a site code and the number of years marshes were isolated from tidal flooding.

Discussion

Our comparisons of reference and recovering diked marshes inform not only restoration activities on the Bay of Fundy, but give a perspective on recovery trajectories of reclaimed salt marshes in general. Our results also reveal interesting characteristics of vegetation zonation on the Bay of Fundy that add to our knowledge, or perhaps challenge our understanding of controls on *S. patens*' distribution and marsh vegetation zonation. Our discussion focuses on these three aspects, beginning with the last mentioned.

Spartina patens

S. patens is primarily found on the western Atlantic and in the northern hemisphere ranging from the subtropics to 48°N (Canadian Museum of Nature Herbarium). Its distribution within Bay of Fundy marshes follows the pattern seen in New England marshes to the south. It is situated at an elevation intermediate to *S. alterniflora* and *J. gerardii*, albeit in the Bay of Fundy other species cooccur with greater abundance than found in the south. Our results reveal a trend of increasing vertical range of *S. patens* with tidal range – a condition recognized for *S. alterniflora* (McKee and Patrick 1988), but not previously reported for *S. patens*.

The vertical range of a species' distribution should be predictable by the hydroperiod and frequency of tidal inundation. Harvey and Odum (1990) indicated that the *S. alterniflora* of the mesotidal Barnstable marsh of Cape Cod, Massachusetts, is inundated more than 675 times per year. Blum (1968) noted that *S. patens* is flooded > 88 times per year in a microtidal marsh of the York River in Virginia. On the Bay of Fundy average inundation frequencies of these species is lower. In some years portions of the *S. patens* platform may be subject to a single flooding event (Table 1). The duration of a tidal flooding event decreases with increasing tidal range (tidal velocity is increased), so hydroperiods are lower for these species in our study area. If the lower elevation of *S. patens* is limited by its tolerance for stresses associated with hydroperiod and its upper limit by competition (as reviewed by Pennings and Bertness 2001), then decreased hydroperiods can explain the greater vertical range of *S. patens* in our macrotidal marsh, but the decreased hydroperiod should also allow it to exist at lower elevations. Our observations suggest that other limitations on the distribution of *S. patens* may exist and further environmental studies at these extreme conditions are warranted.

Rates of Recovery and Marsh Resilience

A number of studies have reported relatively rapid recovery of salt marsh vegetation in nonconstructed marshes. In a Rhode Island marsh, vegetation was converging towards target conditions after a single year (Roman et al. 2002) and in Connecticut typical salt marsh vegetation was established in 10 to 20 yr (Sinicrope et al. 1990; Warren et al. 2002). Although these sites were diked, they were systems that impounded water and had emergent wetland vegetation and small areas of salt marsh species were present when restoration was initiated.

Our study aims to assess the resilience of tidal salt marsh vegetation when allowed to recover after transformation to a managed, terrestrial ecosystem (i.e., reclaimed for agriculture). Examination of trajectories of recovery rates is a measure of the resilience (sensu Grimm and Wissel 1997) of tidal salt marshes and indication of feasible outcomes over the long term. Although recovery should be considered with respect to multiple marsh functions we limit our comparison to plant species' cover and production because vegetation integrates environmental factors and vegetation data has been used to assess recovery from agricultural reclamation in marshes with other tidal regimes and dominant vegetation (Crooks et al. 2002; Thom et al. 2002).

Similar studies have been performed on marshes with different pools of species available for colonization, varying periods of tidal isolation, and varying tidal ranges (Fig. 4). Thom et al. (2002) compared vegetation cover by calculating similarities (weighted and unweighted) between a reference and a recovering marsh to assess recovery of a marsh on Washington's Elk River at 4, 6, and 11 yr after dikes had been breached. We use data from recovering and reference macrotidal salt marshes in England (Crooks et al. 2002) to make the same comparison, but based on end-of-season standing crop. Although dikes were in place for 70 yr at Elk River, the site was dominated by freshwater plants and supported some salt marsh species, likely contributing to the $> 70\%$ similarity to a reference site after just 11 yr of recovery.

On the Bay of Fundy, the Saints Rest marsh was isolated from tidal flooding for approximately 100 yr, but observations from historical aerial photographs suggest that freshwater wetlands existed in upper reaches of the marsh creek and there may have been salt marsh vegetation present just inside the dykes. In 50 yr of recovery similarities are equivalent or slightly higher than those at Elk River at 11 yr. The lower similarities at the John Lusby marsh may be expected due to its longer period of isolation from tidal flooding (300 yr) and recovery period that is only 10 yr longer than the Saint's Rest marsh.

Marshes of the Essex estuaries studied by Crooks et al. (2002) have been diked since medieval times, yet recovering marshes are 82% to 96% similar to a reference site after 100 yr of unassisted recovery. What is most encouraging is that weighted similarities, which reflect the abundance of species, reach the highest values here and in the Bay of Fundy marshes. On this basis we may consider vegetation recovery a success in our two recovering marshes.

The apparent success in Bay of Fundy marsh growth following dike breach is related to several factors, which include the high marsh platform elevation when dikes were breached and large vegetation growth range, as well as rapid rates of sediment deposition. At Saint's Rest marsh the reclamation surface is located at depths of 11 to 21 cm in the *S. patens* dominated zone (Noel et al. 2005), which means that marsh surface elevations were 3.64 to 3.74 m above MSL when the dike breached. Considering the present elevation and a sea level that was 20 cm lower in 1950 (Desplanque and Mossman 2004), the present high marsh surface was probably within the elevation range of *S. alterniflora*. At the John Lusby marsh the reclamation surface is located at 100 to 130 cm below the *S. patens*-dominated zone (Graf 2004), which means that when the dike was breached the marsh surface also was within the elevation range of *S. alterniflora*. The high sediment deposition rates (Chmura et al. 2001) and large growth range of *S. alterniflora* makes Bay of Fundy marshes resilient in the face of diking and ditching and prospects for success of deliberate restoration efforts are promising, without the addition of fill required for marsh restoration in some regions, such as the diked marshes of the Delaware estuary in New Jersey (Weinstein and Weishar 2002).

Morris et al. (2005) hypothesize that the distribution of salt marsh elevation relative to MHW is diagnostic of marsh stability in the face of relative sea-level rise. Specifically, highly stable marshes will have mean elevations approximately equal to MHW and well above MSL or the lower limit of vegetation growth. The Bay of Fundy marshes included in this study all have mean platform elevations at or above MHW and minimum platform elevations are much greater than the lower limit for *S. alterniflora*. These findings suggest that, in the vertical plane, the marshes studied are stable and are not imminently threatened by rising relative sea level.

There is a high probability of success for restoration of the reclaimed salt marshes of the Bay of Fundy. Returning the tides to these agricultural lands could help to offset the marsh loss expected as rising sea levels threaten marshes at lower tidal ranges and marsh migration is prevented by the coastal squeeze of developed uplands that is predominant in marshes along the urbanized U.S. coastline. Presently, many of these reclaimed lands on the Bay of Fundy are becoming sites of commercial and residential development. The potential to restore reclaimed marshes, and reduce habitat loss depends upon prohibition of this kind of land use.

Acknowledgments

We are grateful for the assistance of C. Boleyn, P. Byers, G. Hung, G. MacDonald, A. Sabourin, and P. Zavoda. We thank the staff of the Coastal Wetlands Institute (Mount Allison University) who allowed us to use their facility to process plant samples and the Canadian Wildlife Service for permission to conduct research at John Lusby marsh. This research was financially supported by a (NSERC) awarded to G. Chmura, a McGill Frazee Scholarship (Huntsman Marine Science Centre) awarded to S. Byers, and a summer NSERC awarded to field assistant A. Sabourin.

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Received, February 7, 2007
Revised, July 10, 2007
Accepted, July 17, 2007